STAFF REVIEW · H446 248 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. It discards half the candidates based on ordering.

2. A condition that stops recursive calls.

3. They credit correct intermediate reasoning and algorithm states.

4. Q=1 immediately after first rising edge; remains 1 between edges even though D changes; at next rising edge Q becomes 0.

I Do — source worked steps

1. Predict first — Start with +9 in exactly eight bits and invert it.

2. Explain the answer — Add one to give 11110111, representing −9. This explicitly shows the stages.

We Do — guided CHECK

1. Predict first
Compare both candidate routes, recording their total costs.
Explain the answer
The two-edge route costs 6, which is smaller than the direct route cost 9; explain why.

You Do — source mark schemes

1. [4] Compare 3 [1], 5 [1], 7 [1], then return index 2 (zero-based) [1].

2. [6] Positive 6=00000110 [2]; invert to 11111001 [2]; add one to 11111010 [2]. The claimed value is one less and represents −7.

3. [6] low0 high6 mid3→7 [1]; target larger so low4 [1]; low4 high6 mid5→11 [1]; increase low6 [1]; mid6→13 found [1]; bounds shrink on each unsuccessful step [1].

Exit Ticket — source answer

1. False. Independent success with a different example is better evidence.